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A Phase 3 Biomarker Validation of GALAD for the Detection of Hepatocellular Carcinoma in Cirrhosis

Tracey L. Marsh, Ph.D.¹, Neehar D. Parikh, M.D.², Lewis R. Roberts, M.D.³, Myron E. Schwartz, M.D.⁴, Mindie H. Nguyen, M.D.⁵, Alex Befeler, M.D.⁶, Stephanie Page-Lester¹, Nabihah Tayob, Ph.D.⁷, Sudhir Srivastava, Ph.D.⁸, Jo Ann Rinaudo, Ph.D.⁸, Amit G. Singal, M.D.⁹, K. Rajender Reddy, M.D.¹, Jorge A. Marrero, M.D.¹

¹Biostatistics Program, Public Health Sciences Division, Fred Hutchinson Cancer Research Center, Seattle, WA

²Division of Gastroenterology and Hepatology, University of Michigan, Ann Arbor, MI

³Division of Gastroenterology and Hepatology, Mayo Clinic College of Medicine and Science, Rochester, MN

Correspondence: Jorge A. Marrero, M.D., Division of Gastroenterology and Hepatology, University of Pennsylvania Perelman School of Medicine, Perelman Center for Advanced Medicine, 3400 Civic Center Blvd, 713 South Pavilion, Philadelphia, PA 19104, Fax: 215-893-5570, jorge.marrero@pennmedicine.upenn.edu.

Authorship contributions:

Tracey L. Marsh, PhD: (Formal analysis: Lead; Data curation: Lead ; Methodology: Lead; Writing — original draft: Co-Lead; Writing — review & editing: Co-Lead)

Neehar D. Parikh, MD: (Conceptualization: Collaborator; Formal analysis: Collaborator; Methodology: Collaborator; Writing — original draft: Collaborator; Writing — review & editing: Collaborator)

Lewis R. Roberts, MD: (Conceptualization: Collaborator; Formal analysis: Collaborator; Methodology: Collaborator; Writing — original draft: Collaborator; Writing — review & editing: Collaborator)

Myron E. Schwartz, MD: (Conceptualization: Collaborator; Formal analysis: Collaborator; Methodology: Collaborator; Writing — original draft: Collaborator; Writing — review & editing: Collaborator)

Mindie H. Nguyen, MD: (Conceptualization: Collaborator; Formal analysis: Collaborator; Methodology: Collaborator; Writing — original draft: Collaborator; Writing — review & editing: Collaborator)

Alex Befeler, MD: (Conceptualization: Collaborator; Formal analysis: Collaborator; Methodology: Collaborator; Writing — original draft: Collaborator; Writing — review & editing: Collaborator)

Stephanie Page-Lester: (Data curation: Co-Lead; Data Coordination and Management- Co-Lead)

Nabihah Tayob, PhD: (Data curation: Collaborator; Statistical Analysis: Collaborator; Writing — Collaborator; Writing — review & editing: Collaborator)

Sudhir Srivastava, PhD: (Conceptualization: Co-Lead; Funding acquisition: Lead; Investigation: Lead; Methodology: Lead; Supervision: Lead; Writing — original draft: Collaborator; Writing — review & editing: Collaborator)

Jo Ann Rinaudo, PhD: (Conceptualization: Co-Lead; Funding acquisition: Program Director; Methodology: Lead; Writing — Collaborator; Writing — review & editing: Collaborator)

Amit G. Singal, MD: (Conceptualization: Collaborator; Formal analysis: Collaborator; Methodology: Collaborator; Writing — original draft: Collaborator; Writing — review & editing: Collaborator)

K. Rajender Reddy, MD: (Conceptualization: Collaborator; Formal analysis: Collaborator; Methodology: Collaborator; Writing — original draft: Collaborator; Writing — review & editing: Collaborator)

Jorge A. Marrero, MD: (Conceptualization: Lead PI; Formal analysis: Co-Lead; Data curation: Lead; Funding acquisition: Lead; Investigation: Lead; Methodology: Lead; Visualization: Lead; Supervision: Lead; Writing — original draft: Co-Lead; Writing — review & editing: Co-Lead).

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⁴Recanati/Miller Transplant Institute, Mt Sinai Medical Center, New York, NY

⁵Division of Gastroenterology and Hepatology and Department of Epidemiology and Population Health, Stanford University, Palo Alto, CA

⁶Division of Gastroenterology, Saint Louis University, St. Louis, MO

⁷Department of Data Science, Dana Farber Cancer Institute, Boston, MA

⁸Cancer Biomarkers Research Group, Division of Cancer Prevention, NCI, Bethesda, MD

⁹Division of Digestive and Liver Disease, UT Southwestern, Dallas, TX

¹⁰Division of Gastroenterology and Hepatology, University of Pennsylvania, 3400 Civic Center Blvd, 713 South Pavilion, Philadelphia, PA 19104

Abstract

BACKGROUND & AIMS: Better surveillance tests for hepatocellular carcinoma (HCC) are needed. The GALAD score [Gender, Age, AFP-L3, AFP, and Des-carboxy-prothrombin] has been shown to have excellent sensitivity and specificity for HCC in phase two studies. We performed a phase three biomarker validation study to compare GALAD with AFP in detecting HCC.

METHODS: This is a prospective study of patients with cirrhosis enrolled at seven centers. Surveillance for HCC was performed every 6 months at each site, and HCC diagnosis was confirmed per AASLD guidelines. Blood for biomarker research was obtained at each follow-up visit and stored in a biorepository. Measurements of AFP, AFP-L3, and DCP (des-gamma carboxyprothrombin) were performed in a FujiFilm laboratory by staff blinded to clinical data. The performance of GALAD in detecting HCC was retrospectively evaluated within 12 months prior to the clinical diagnosis. All analyses were conducted by an unblinded statistician in the EDRN data management and coordinating center.

RESULTS: A total of 1,558 patients with cirrhosis were enrolled and followed for a median of 2.2 years. A total of 109 patients developed HCC (76 very early or early stage) with an annual incident rate of 2.4%. The AUC for AFP and GALAD within 12 months prior to HCC 0.66 and 0.78 ($p < 0.001$), respectively. Using cutoff for GALAD of -1.36 , the specificity was 82% and sensitivity at 12 months prior to HCC diagnosis was 62%. For comparison, performance of AFP at 82% specificity showed 41% sensitivity at 12 months prior to HCC diagnosis ($p = 0.001$).

CONCLUSION: GALAD score, compared to AFP, improves the detection of HCC within 12 months prior to the actual diagnosis.

Lay Summary

Better surveillance tests for hepatocellular carcinoma, primary liver cancer, are needed. We perform a large prospective study to compare alpha-fetoprotein to GALAD for the detection of liver cancer. We showed that GALAD was more sensitive and specific for the diagnosis of liver cancer in patients with liver cirrhosis.

Keywords

early detection; hepatocellular carcinoma; GALAD; AFP

INTRODUCTION

Hepatocellular carcinoma (HCC) is the third leading cause of cancer-related mortality worldwide and a leading causes of death in patients with liver cirrhosis.¹ However, prognosis is driven by tumor stage, with curative options available only if patients are detected at an early stage.² The close association between early detection and improved survival underlies the recommendation of HCC surveillance in patients with cirrhosis by multiple professional societies.^{3,4} HCC surveillance is performed using semiannual abdominal ultrasound with or without alpha-fetoprotein (AFP); however, several limitations highlight a need for new surveillance strategies. The current surveillance tests performance characteristics may have been over estimated due to the lack of large scale validation studies.⁵ Ultrasound-based surveillance can lead to screening-related harms due to false-positive or indeterminate screening results.^{6,7} Finally, utilization of surveillance for HCC in patients with cirrhosis has been shown to be around 25%, related to a multitude of patient-reported barriers.^{8,9} Overall, there is a clear need for more effective HCC surveillance tests to improve detection of HCC and reduce HCC-related mortality.

Several serum-based biomarkers and biomarker panels have promising data for HCC detection in patients with cirrhosis.¹⁰ AFP remains the only serum biomarker to undergo all five phases of biomarker validation for HCC surveillance.² Multicenter data indicate that GALAD (gender, age, AFP-L3%, AFP and, des-gamma-carboxy prothrombin [DCP]) can achieve higher sensitivity for early HCC detection, exceeding 70%, in case-control studies,^{11,12} and a small prospective study showed sensitivities of 52% and 72% for AFP and GALAD, respectively, when analyzed six months prior to HCC development.¹³ Although these data are promising, an expert panel and guidelines have stressed the critical importance of large, adequately powered longitudinal cohort studies (phase three biomarker validation) to determine if biomarkers can detect cancer early before it becomes clinically evident.¹⁴ The NCI's Early Detection Research Network launched the Hepatocellular carcinoma Early Detection Strategy (HEDS) cohort to establish a large biorepository of prospectively collected blood samples and a clinical database from patients at high risk for developing HCC.¹⁵ The first biomarker validation study utilizing HEDS is this evaluation of GALAD for detecting HCC in patients with cirrhosis.

METHODS

Design.

We performed a large multicenter prospective phase three biomarker validation study following recommendations by the National Cancer Institute's Early Detection Research Network¹⁶ and expert guidelines.¹⁷ The details of the HEDS study design can be found elsewhere.¹⁵ The study was approved by each site's institutional review boards.

Study Population.

Consecutive patients with cirrhosis were enrolled at 7 sites in the United States: University of Pennsylvania, University of Michigan, UT Southwestern Medical Center, Stanford University, Mayo Clinic Rochester, Saint Louis University and Mt. Sinai Medical Center.

As previously described,¹⁵ the inclusion criteria were: a) age > 18 years of age, b) diagnosis of cirrhosis, c) preserved liver function with Child class A or B and model for end stage liver disease (MELD) < 15, d) ultrasound (US), CT or MRI showing no liver mass in the previous 6 months, and f) dynamic CT scan or triple phase MRI demonstrating lack of a liver mass within 3 months prior to enrollment for those with AFP > 20 ng/ml at baseline. Exclusion criteria include clinical evidence of significant hepatic decompensation (such as refractory ascites, grade 3 or 4 hepatic encephalopathy, hepatorenal syndrome, Child class C), significant medical comorbidities with life expectancy < 1 year, cancer history within the last five years, need for long-term immunosuppression, and prior solid organ transplant.

Study Procedures.

Patients were identified and informed consent was obtained. Baseline data collected included demographic data, medical history including liver disease etiology; body mass index (BMI), and presence/absence of ascites and any grade 3 or 4 hepatic encephalopathy; routine hepatic biochemical tests, complete blood count, INR; and research blood sample. Chronic liver disease was categorized as follows: 1) Hepatitis B (HBV) infection, 2) Hepatitis C (HCV) infection, 3) alcoholic liver disease (ALD), 4) metabolic associated steatotic liver disease (MASLD), or 5) other [cryptogenic (if no features of metabolic syndrome), autoimmune hepatitis, primary biliary cholangitis, or primary sclerosing cholangitis]. Patients with multiple etiologies were assigned to the first applicable category in this ordered hierarchy. Liver disease severity was assessed using Child Pugh classification and the MELD score.

Surveillance testing was performed following guidelines for the standard of care for patients with cirrhosis.⁴ Patients were followed every six months per local site standards, including abdominal imaging test (US, MRI or CT scan), AFP, hepatic biochemical tests, and complete blood counts. A research blood sample was obtained at each follow up, and any changes in medical history were recorded. The processing of the blood samples was performed as previously described using NCI's Early Detection Research Network standards.¹⁵

HCC was defined using American Association for the Study of Liver Diseases HCC guidance,⁴ and tumor burden stage for HCC was defined as Barcelona Clinic Liver Cancer (BCLC) staging, simplified to use tumor number, size, and invasiveness but not performance status or Child class. Very early and early stage correspond to BCLC stages 0 and A, respectively, and herein are collectively referred to as Early HCC. The date of diagnosis used as a benchmark for early detection is the date of HCC confirmation. For post hoc analysis, the imaging modality, either US or MRI/CT, leading to the diagnostic workup and confirmation of HCC was assigned by consensus between study investigators J.A.M. and A.G.S. after review of the imaging records for each patient diagnosed with HCC.

Assays.

Serum from participants were identified by the Data Management and Coordinating Center in January 2020, and transferred to Fujifilm Wako Diagnostics lab for AFP, AFP-L3, and DCP measurements. Fujifilm Wako Diagnostics were blinded to clinical data and patient

status (HCC cases or cirrhosis controls). AFP, AFP-L3%, and DCP were performed using a microchip capillary electrophoresis and liquid-phase binding assay on a μ TASWako i30 auto analyzer (Wako Pure Chemical Industries, Ltd., Osaka, Japan). GALAD was calculated using the following equation: $Z = -10.08 + (0.09 \text{ age}) + (1.67 \text{ male sex}) + (2.34 \log \text{ AFP}) + (0.04 \text{ AFP-L3}) + (1.33 \log \text{ DCP})$.¹¹ The clinical database used in this analysis was archived in May 2022, participants with assayed specimens were consented between April 2013 and December 2018. All analyses were conducted by an unblinded statistician in the EDNRN data management and coordinating center.

Statistical Analysis.

The GALAD model and AFP were evaluated using time dependent receiver operating characteristic (ROC) curve analyses. For each preclinical period, biomarkers from specimens collected within the specified time prior to diagnosis of and up to one month after diagnosis for HCC cases and from all specimens for all cirrhotic controls contributed to analyses. Sensitivity and specificity corresponding to tests at published cutoffs of -1.36 for GALAD¹² and 20 ng/mL for AFP¹⁸ were assessed. For AFP at cutoff 6.5 ng/mL , selected post hoc to yield the same specificity as GALAD with -1.36 cutoff, was also assessed. To summarize overall discrimination, ROC curves were plotted and the area under the ROC curve (AUC) was estimated. Additionally, we evaluated DCP as single biomarker and sub-models of GALAD defined by calculating scores using the subset of coefficients corresponding to the included biomarkers. For example: the subset of gender, age, and AFP (GAA) is calculated using the equation: $Z = -10.08 + (0.09 \text{ age}) + (1.67 \text{ male sex}) + (2.34 \log \text{ AFP})$. Other subsets, such as GAD (gender, age, and DCP) and GAAD (GALAD without AFP-L3%), are calculated similarly. We evaluated AUC and the sensitivity or specificity corresponding to tests with cutoffs selected to respectively target sensitivity or sensitivity of 90% and to target the post hoc scenario of 82% specificity. Performance characteristics were also evaluated for detection of early HCC and separately for patients with viral and non-viral etiology of liver disease. For a post-hoc analysis to explore the impact of surveillance modality on sensitivity, analyses with specificity fixed at 90% were repeated separately for HCC cases diagnosed while under apparent surveillance with US or MRI/CT leading to HCC detection. For all estimates, 95% confidence intervals (CI) were calculated as percentiles from 2,000 bootstrap resamples, as were two-sided p values for comparisons. Clustered resampling, conditional on the number of case and control patients in each analysis, was employed to account for correlation between multiple measurements within patients.^{19,20} To limit the possibility of undiagnosed HCC cases being incorrectly evaluated as controls in the biomarker evaluations (i.e., ascertainment bias), specimens collected within 6 months of non-HCC patient's last clinical, imaging, or laboratory visit were excluded from analyses. Biomarker values from specimens collected more than one month after diagnosis were excluded from all analyses. Throughout, we report results for the potential of tests for detection, within 3, 6, and 12 months prior to diagnosis of HCC in tables and focus in the text on detection within 12 months prior to diagnosis of HCC. The focus on 12 months reflects a higher bar for biomarkers when evaluating their potential for early detection and highlights estimates based on greater numbers of patients with HCC diagnosis.

RESULTS

Patient Characteristics.

In the HEDS cohort, a total of 109 patients developed HCC after a median follow up of 2.2 years (range:0-8.7), yielding an annual HCC incidence rate of 2.4%.²⁰ This study selected available serum samples from 1,558 prospectively enrolled patients. Of these, 1,123 patients without HCC and with at least 6 months follow-up after serum collection and 107 HCC cases had biomarkers included in analysis (Figure 1). Among the case patients, 69, 56, and 33 had at least one specimen collected within 12, 6, and 3 months prior to diagnosis, respectively. Table 1 shows baseline characteristics of patients in the biomarker cohort. To summarize, for the cirrhosis controls, 51.6% were males, median age was 60 (interquartile range (IQR):54,65), and median BMI was 30 (IQR:26.4,35.2). Most of the patients were non-Hispanic White (80.6%), with 9.2% being Hispanic and 6.9% Black. The most common liver disease etiology was chronic HCV (42.4%), followed by MASLD (26.3%) and ALD (14.7%). Most patients had compensated Child class A cirrhosis (73.9%), and the median MELD score was 8 (IQR: 7-11). Of the 107 patients who developed HCC, the tumor staging was very early in 21 (19.6%), early in 55 (51.4%), and late in 31 (29%).

Performance Characteristics of GALAD and AFP.

The distribution of the biomarkers was determined for all control specimens and case specimens collected within 12 months prior to diagnosis (Supplementary Figure 1). The median GALAD values were -2.64 (IQR: $-3.63, -1.70$) and -1.47 (IQR: $-2.52, -0.56$) for controls and cases, respectively with a difference in medians of 1.17 (95% CI: 1.03,1.48). The median AFP values were 3.50 ng/ml (IQR: 2.40,5.30) and 5.40 ng/ml (IQR: 3.20,8.70) for controls and cases, respectively, with a difference in medians of 1.9 (95%CI:1.65,2.7) ng/mL. Performance characteristics of the tests derived from GALAD and AFP using previously published cutoffs are illustrated in Figure 2 and reported in Table 2. Paired with the cutoff of -1.36 (original manuscript¹²), GALAD demonstrated specificity of 82% and 62% sensitivity within 12 months prior to HCC diagnosis. For comparison, performance of AFP at 82% specificity, cutoff 6.45 ng/ml (95% CI: 5.95,6.85), showed 42% sensitivity within 12 months prior to HCC diagnosis ($p=0.001$). The currently recommended cutoff for AFP (20 ng/ml), was highly specific (97% specificity) and sensitivity dropped notably with increasing lead time (24% within 3 months and 12% within 12 months prior to diagnosis). The ROC curves for GALAD and AFP within 3, 6 and 12 months prior to the diagnosis of HCC are shown in Figure 2 with corresponding AUC reported in Table 3. The AUC within 12 months prior to HCC diagnosis was 0.78 for GALAD and was 0.66 for AFP ($p<0.001$).

When restricted to Early HCC, 12 month sensitivities at common 82% specificity were 60% for GALAD (cutoff -1.65) and 42% for AFP (cutoff 6.56 ng/mL) and was 11% for AFP with cutoff 20 ng/mL (97% specificity). The AUCs within 12 months prior to HCC diagnosis were 0.77 for GALAD and 0.64 for AFP. HCC stage and detailed patient and specimen counts for time-dependent analyses are reported for preclinical periods of up to three years in Supplementary Tables 1–6.

Post hoc exploration.

Of the 107 HCC case patients reviewed, a total of 51 patients with incident HCC appeared to have undergone an US-based surveillance strategy, while 56 appeared to have undergone a CT/MRI based surveillance strategy (Supplementary Tables 1,6–7). At 82% specificity, GALAD (cutoff -1.36) is more sensitive than AFP (cutoff 6.45) for detection within 1-year of diagnosis, under either surveillance modality. Among MRI, sensitivities 57% vs 34% ($p=0.006$) and among US, sensitivities 68% vs 50% ($p=0.04$), were observed for GALAD vs AFP. The differences in sensitivities for within 3 and within 6 months of diagnosis are similar and reported in Supplementary Table 7.

Performance Characteristics of DCP and GALAD Sub-models.

In addition to AFP (A) and GALAD, we consider DCP (D) as a single biomarker and combinations of gender and age with AFP (GAA), or DCP (GAD), or both (GAAD). ROC curves (Figure 3), AUC values (Table 3), specificity at 90% sensitivity and sensitivity at 82% specificity (Figure 3, Table 4, corresponding cutoffs in Supplementary Table 9), collectively suggest incremental contributions of biomarkers to early detection. For overall discrimination within 12 months of HCC diagnosis, AFP and DCP demonstrated AUCs of 0.66 and 0.71, respectively, combinations with gender and age demonstrated AUCs of 0.71 (GAA) and 0.74 (GAD), respectively. The AUC for gender and age combined with both AFP and DCP (GAAD) was 0.76 and was 0.78 for the full GALAD model. For cutoffs yielding 90% sensitivity to detect HCC within 12 months prior to diagnosis, specificity of AFP and DCP were 18% and 25%, respectively and the addition of age and gender yielded 18% and 24% specificity, respectively. At this high sensitivity, combinations with gender, age, and both AFP and DCP demonstrated 35% specificity, with or without AFP-L3% (GAAD and GALAD). At 82% specificity, the sensitivity of AFP and DCP within 12 months of HCC diagnosis was 42% and 50%, respectively and inclusion of age and gender demonstrated 44% and 52% sensitivity for GAA and GAD, respectively. Gender and age combined with both AFP and DCP (GAAD) demonstrated 57% sensitivity and inclusion of AFP-L3% (the full GALAD model) demonstrated 62% sensitivity. The sensitivities at 90% specificity demonstrated a similar pattern as the 82% specificity scenario with lower sensitivities: AFP and DCP showed 27% and 28% sensitivity, respectively, GAA and GAD showed 35% and 39% respective sensitivities, and 42% and 47% sensitivity were respectively demonstrated by GAAD and GALAD (Supplementary Table 8).

Subgroup analysis.

The performance characteristics of GALAD and sub-models of GALAD were evaluated in subgroups of HCC cases and cirrhosis controls with viral and non-viral etiologies (Table 3). There were 58 HCC and 493 cirrhosis controls with viral etiologies and 49 HCC cases and 630 cirrhosis controls with non-viral etiologies; 36 and 33 HCC cases had specimens within 12 months prior to diagnosis in the viral and non-viral group, respectively. Among viral etiology, all HCC cases and 94% of cirrhosis controls had HCV without HBV etiology. GALAD with cutoff of -1.36 demonstrated 82% specificity in the full cohort, 81% specificity among viral, and 83% specificity among non-viral; sensitivity for within 12 months of HCC diagnosis was 74% among viral and 49% among non-viral. AFP with

cutoff of 6.45 ng/mL, selected to target 82% specificity in the full cohort, demonstrated 71% specificity among viral and 92% specificity among non-viral. AFP paired with cutoffs of 9.53 ng/mL and 4.95 ng/mL respectively targeted 82% specificity in each group and showed 35% and 39% sensitivity among viral and non-viral groups, respectively. DCP paired with cutoffs of 0.64 mAU/mL and 2.07 mAU/mL respectively targeted 82% specificity in each group and showed 64% and 41% sensitivity among viral and non-viral groups, respectively. At both 82% and 90% specificity, the sensitivity within 12 months of HCC diagnosis was lower in the non-viral group than the viral group for every sub-model of GALAD considered except AFP, as noted above. Also, the cutoffs required to target 82% and 90% specificity were lower in the non-viral group than the viral group for every sub-model of GALAD considered except DCP and GAD. This pattern was also demonstrated by AUCs across viral and non-viral groups (Tables 3 & 4, Supplementary Table 9).

DISCUSSION

To the best of our knowledge, our study is the first multicenter prospective biomarker phase 3 study that evaluates AFP and GALAD for the detection of HCC among patients with cirrhosis. Among the 109 patients who developed HCC, a median follow-up of 2.2 years and planned biannual specimen collection allowed sensitivity to be examined over a range of preclinical periods. The test from AFP with a 20 ng/mL cutoff was highly specific (97% specificity) with low sensitivity (12%-24% between 12 and 3 months prior to diagnosis). When comparing GALAD with cutoff -1.36 to AFP at a common 82% specificity, GALAD showed 19-21% absolute increases in sensitivity over AFP for detecting HCC within 3, 6, and 12 months of diagnosis (respective p-values: 0.02, 0.003, 0.001). Overall discrimination between patients with cirrhosis and patients within 12 months prior to HCC diagnosis was assessed by AUC of 0.78 for GALAD and 0.66 for AFP ($p < 0.001$); performances were similar for patients within 3 and 6 months prior to HCC. Our study showed that GALAD had better performance characteristics over AFP in terms of AUC and sensitivity at 90% and 82% specificity. GALAD also showed potential to reduce more surveillance MRI or CT scans than AFP while retaining 90% sensitivity for referring MRI/CT scans to patients within 3, 6, and 12 months prior to HCC development. However, the optout potential of GALAD requires further consideration of specificity and sensitivity trade-offs and confirmation in a study with greater frequency of labs among HCC case patients in the lead up to diagnosis.

The GALAD score is made up of 3 biomarkers together with age and gender. The optimal cutoff is unknown. The GALAD threshold of -1.36 was developed in the original report,¹² to optimize sensitivity at 80% specificity. In our study, this cutoff was validated for specificity (82% specificity observed) and demonstrated sensitivities 62-65% for within 3, 6, 12 months of the diagnosis of HCC. Interestingly, the specificity of GALAD with this cutoff was robust across viral and non-viral group whereas AFP (with either 20ng/mL or 6.45 ng/mL cutoff) was not. In an earlier EDRN sponsored case-control study, conducted prior to widespread attainment of SVR among treated HCV, AFP with cutoff 20 ng/mL demonstrated 90% specificity.¹⁸ At that time there was significant concern about burden on healthcare systems from advanced imaging. As observed in this study, utilization of advanced imaging may have become more common. For these reasons, in addition to the

high mortality associated with HCC we prioritized sensitivity for early detection and focused on tests with 82% specificity and reported analogous results for tests with 90% specificity.

While this study was not designed to examine the incremental contributions of each component in the GALAD combination, our evaluation of DCP and sub-models of GALAD suggested incremental utility for each component. Of note, DCP and GAD (DCP combined with age and gender) generally exhibited improved performance over AFP and GAA (AFP combined with age and gender), respectively. The complementary aspect of AFP and DCP appeared robust as GAAD demonstrated increased performance over combinations with either AFP or DCP. Additionally, the contribution of AFP-L3% to GALAD was demonstrated by a 5-percentage point increase in sensitivity for detection within 12 months of HCC diagnosis using test cutoffs that target 90% or 82% specificity. A 3-5 percentage point greater sensitivity of GALAD over GAAD persisted for detection within 24 and 36 months of diagnosis. Though the difference in AUC between GALAD and GAAD is small (0.78 vs. 0.76), the ROC curves illustrate that this difference follows from increased sensitivity in the high specificity region. This directly reflects the characteristics of AFP-L3%, a highly specific biomarker that can detect a subset of HCC cases and its consequent contribution to a biomarker combination.

The sub-models of GALAD generally demonstrated performance drops in the non-viral setting relative to the viral setting. Exceptions are for AFP and GAA, and for sensitivity of DCP at 82% specificity. Sample sizes are small in the subset analyses and the incremental improvements in sensitivity from combining age and gender with either AFP, DCP, or both using coefficients from the GALAD model were unclear in the viral subgroup. These need further study as the epidemiology of the underlying liver disease leading to HCC changes towards a predominantly non-viral etiology. For all biomarkers, notable differences in cutoffs were required to target a set specificity in the viral (96% HCV) relative to the non-viral setting. Cutoffs for surveillance tests should be evaluated separately for viral and non-viral settings and possibly for smaller subgroups of common etiology.

Surveillance of patients with cirrhosis has been recommended by several societies, and ultrasound, with or without AFP, has been the main surveillance modality. With the estimated utilization of imaging-based surveillance for HCC being about 24%,⁸ our study is of importance because a biomarker-based surveillance may increase adherence and utilization, which in turn may improve outcomes.²¹ A high number of incident HCC were diagnosed at early stages with about 50% appearing to have undergone MRI/CT surveillance, for which post hoc analyses showed reduced sensitivity of both AFP and GALAD. This is expected as MRI tends to detect HCC earlier than ultrasound and is not the general clinical scenario we envision GALAD will be used. Sensitivities observed in our cohort are notably lower than previously reported, which can be further explained by the fact that previous results were from case-control studies^{11,12} with known biases. We evaluated tests based on specimens collected within 3, 6 and 12 months prior to diagnosis, rather than at diagnosis. A small prospective study evaluated GALAD for the detection of HCC and found a sensitivity and specificity of 72% and 85%, respectively, within 6 months prior to the diagnosis of HCC.¹³ Our study has consistent findings but differs in that it has a

larger sample size, a cohort with mostly treated HCV and about 40% fatty liver disease (both alcohol and MASLD), and with a larger proportion of early HCC.

GALAD has been developed and validated for the detection of HCC in several populations that included non-viral²² and viral etiologies of liver disease.²³ In our study, we showed that GALAD improved performance characteristics for early detection compared to AFP in a cohort of mixed viral (45%) and non-viral (55%) etiologies of liver disease. With 81% of the HCVs in the cohort treated for HCV, this may have minimized active HCV viremia's influence in the study.²⁴ Ultrasound has been hampered by poor performance in non-viral etiology, independent of BMI.²⁵ The performance drop observed among non-viral compared to viral etiology shows a need for continued biomarker development for surveillance of HCC in populations of non-viral etiology. Future studies should evaluate longitudinal GALAD using the parametric empirical Bayes algorithm, as it may eliminate the difference in performance seen in non-viral etiology.²⁶

There are limitations to our study.

A significant number of patients appeared to have undergone surveillance with MRI, a common practice especially in transplant centers. Although a study reported high test performance of MRI for HCC surveillance in patients with HBV-related cirrhosis,²⁷ this strategy has not been recommended due to concerns of cost-effectiveness, radiologic capacity, lack of large studies indicating efficacy, and patient preferences when repeated at regular intervals. Together with the high proportion of patients with early-stage HCC, including many with very early stage HCC, presents a high bar for a phase 3 biomarker validation study. While our study is a large multi-center phase 3 biomarker study, it includes a majority Caucasian population, so GALAD should still undergo further validation in other populations. While our study is the largest reported biomarker phase 3 study in HCC, the number of cases is still small for subgroup analysis. Nevertheless, our subgroup analysis findings provide leads for further investigation (e.g., viral versus non-viral). A major limitation of our study is that the conceptualization and design of this cohort predates appreciation of the impact that direct-acting anti-viral therapies would have on achieving widespread HCV cure. A preliminary coarse analysis suggests that different cutoffs to target a common specificity are needed for testing among cured and active HCV infections, but this study was not designed or powered to answer this question. We have utilized a prospective specimen collection with retrospective biomarker evaluation (PRoBE) study design suggested by leading cancer institutes, societies and experts.^{28,14} Specimens collected from patients were retrospectively assayed for AFP, DCP, and AFP-L3% by a laboratory blinded to outcome and clinical data. The longitudinal study design of the HEDS biorepository allows for evaluation of performance characteristics for tests conducted prior to diagnosis which is central for assessing potential for early detection.

In summary, these data suggest that each component of GALAD contributes to early detection. In this phase 3 biomarker validation study, we demonstrated that GALAD leads to significantly higher sensitivity for HCC detection than AFP. These data have supported the conduct of a phase 4 randomized biomarker study to examine clinical outcomes,

including stage migration, when GALAD is used in surveillance ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT06084234) ID NCT06084234).

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Conflicts of interest:

These authors disclose the following: Tracey L. Marsh has no conflicts to disclose. Neehar D. Parikh is a consultant for Exact Sciences, Eli Lilly, and Freenome; has served on advisory boards of Genentech, Eisai, Bayer, Exelixis, Gilead, Wako/Fujifilm; and has received research funding from Bayer, Target RWE, Exact Sciences, Genentech and Glycotest. Lewis R. Roberts is a consultant for Glycotest. Mindie H. Nguyen has consulted and/or served on the advisory board for Intercept, Exact Science, Gilead, GSK, Exelixis; and has received research support (to Stanford University) from Pfizer, Enanta, Astra Zeneca, GSK, Delfi, Innogen, Exact Science, CurveBio, Gilead, Vir Biotech, Helio Health, Glycotest, B.K. Kee Foundation. Alex Befeler has served on the speaker's bureau for Mallinckrodt; and has received research support (paid to Saint Louis University) from Exact Sciences. Amit G. Singal is a consultant with Bayer, FujiFilm Medical Sciences, Exact Sciences, Universal Dx, Glycotest, Roche, Freenome, GRAIL, Genentech, AstraZeneca, Eisai, and Exelixis. K. Rajender Reddy is an advisor to Mallinckrodt, Novo Nordisk, Spark Therapeutics, BioVie; has received research support (Paid to the University of Pennsylvania) from Mallinckrodt, Sequana, Grifols, Exact Sciences, BMS, Intercept, HCC-TARGET, NASH-TARGET; and has served on the data safety monitoring board for Novartis. Jorge A. Marrero has been a consultant for Glycotest (terminated); and has served on the data safety monitoring board of Astra Zeneca.

Data availability

The data that support the findings of this study belong to a biorepository intended for continued blinded validation of biomarkers approved for access to specimens and will not be made available. For information regarding the application process for specimens from HEDS, please see <https://edrn.nci.nih.gov/data-and-resources/specimen-reference-sets/heds-pecimen-reference-sets/>

Abbreviations

AFP	alpha-fetoprotein
ALD	alcoholic liver disease
AUC	area under the curve
BCLC	Barcelona Clinic Liver Cancer
BMI	body mass index
CT	computed tomography scan
DCP	des-gamma-carboxy prothrombin

DLR	diagnostic likelihood ratios
GALAD	Gender, Age, AFP-L3, AFP, Des-carboxy-prothrombin
HBV	hepatitis B
HCC	hepatocellular carcinoma
HCV	hepatitis C
HEDS	Hepatocellular carcinoma Early Detection Strategy
INR	international normalized ratio
MELD	model for end stage liver disease
MRI	magnetic resonance imaging
MASLD	metabolic associated steatotic liver disease
ROC	receiver operating characteristic
STARD	standards for reporting of diagnostic accuracy studies

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WHAT YOU NEED TO KNOW

BACKGROUND AND CONTEXT

Most patients with hepatocellular carcinoma have liver cirrhosis. Current surveillance tests for patients with cirrhosis demonstrate substandard sensitivity and specificity.

NEW FINDINGS

Our study proposed a novel cirrhosis surveillance strategy using GALAD that shows greater sensitivity and specificity.

LIMITATIONS

As a biomarker phase 3 study, this study does not compare GALAD to ultrasound plus AFP in the detection of early stage HCC.

CLINICAL RESEARCH RELEVANCE

Our surveillance model using GALAD can significantly improve early HCC detection prior to the clinical diagnosis of HCC. This study is important as it sets up a clinical utility study that compares GALAD versus ultrasound plus AFP.

BASIC RESEARCH RELEVANCE

This study provides a large research sample set for which novel biomarkers, including liquid biopsy techniques, can be validated for the detection of HCC.

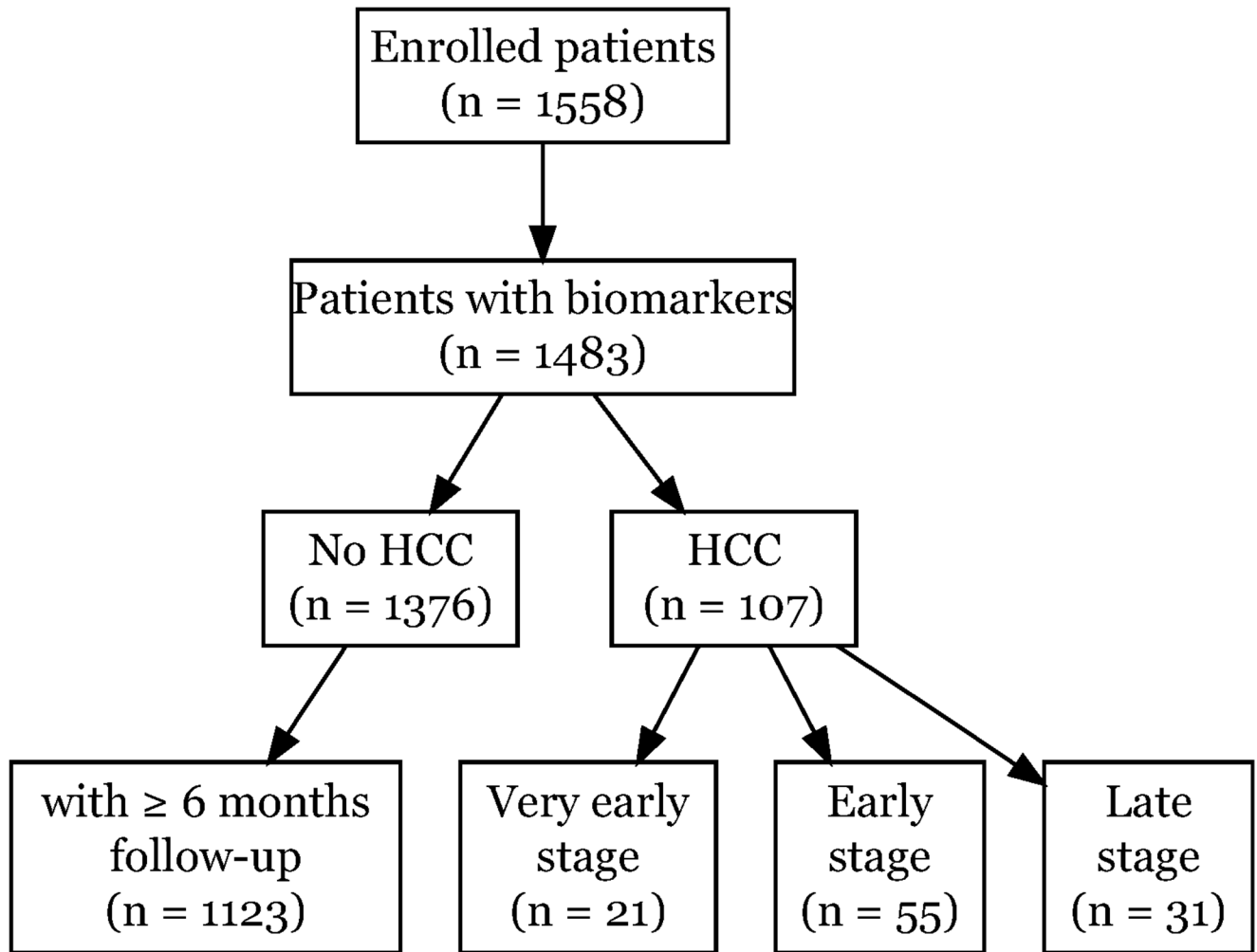


Figure 1.

STARD* diagram to report flow of participants through HEDS study.

Note: 5 patients without HCC had some specimens with measurements for AFP but none with all three GALAD biomarkers. See Supplementary Table 2 for detailed sample sizes. Enrolled patients as of January 2020 when specimens were identified for shipment to the laboratory and number of HCC cases diagnosed as of clinical database snapshot in May 2023. Among patients without HCC, only biomarkers with at least 6 months of post serum collection follow-up were included in analysis.

*STARD – Standards for Reporting of Diagnostic Accuracy Studies

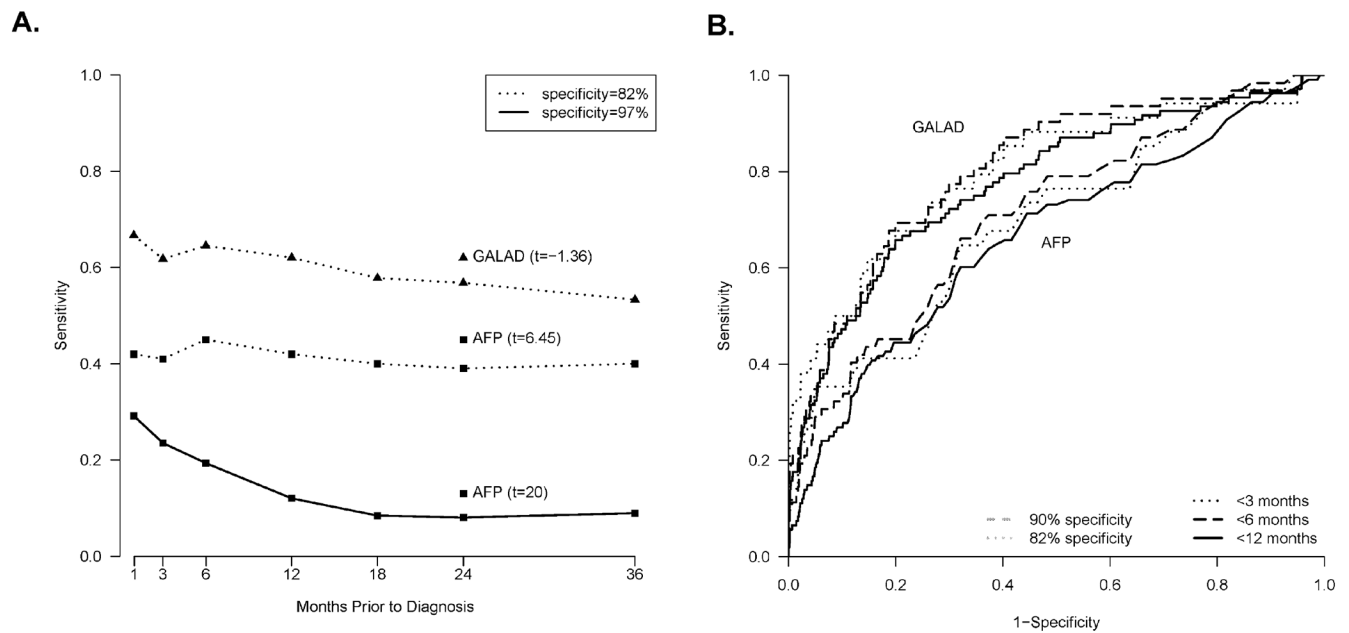


Figure 2.

(A) Performance of rules derived from GALAD and AFP for prediction of HCC within 1 – 36 months. Rules are defined by pairing the biomarkers with published thresholds of $t=-1.36$ for GALAD and $t=20$ for AFP. The sensitivity of AFP paired with $t=6.45$ was included for direct comparison with GALAD ($t=-1.36$) since both demonstrated 82% specificity in the HEDS cohort. **(B)** Receiver Operating Characteristics Curves for the prediction of HCC within 3, 6, and 12 months prior to the diagnosis of Hepatocellular Carcinoma based on GALAD or AFP.

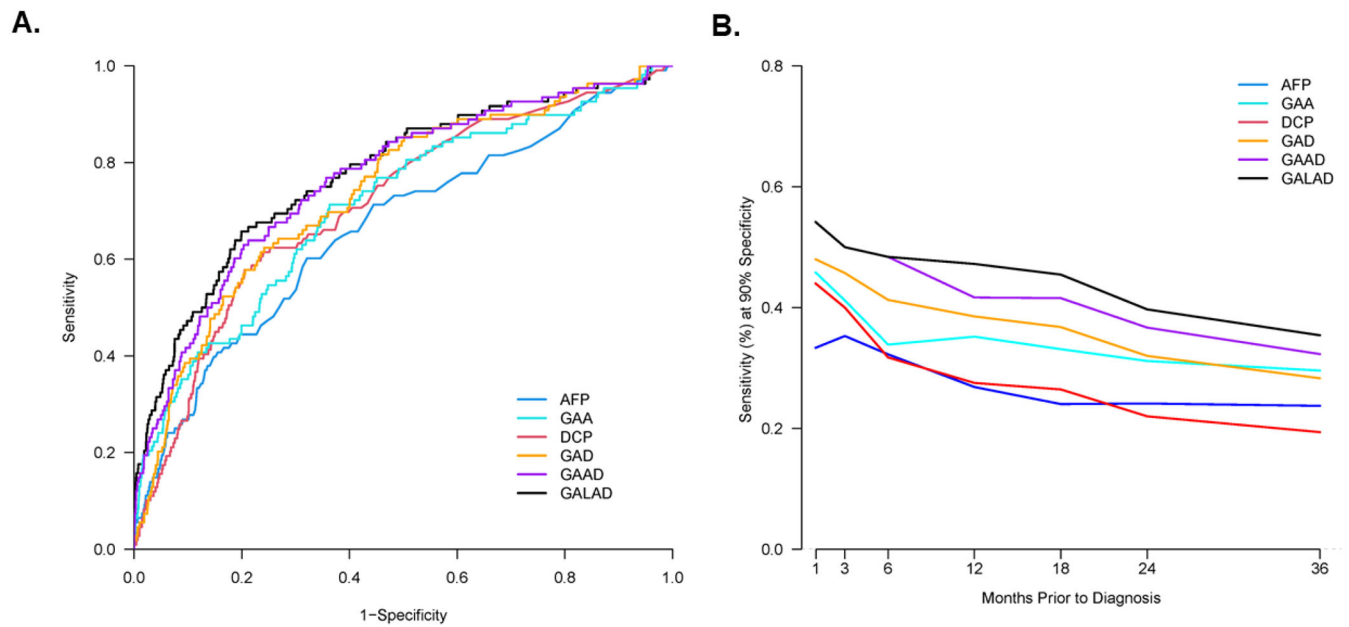


Figure 3.

Performance characteristics for the prediction of HCC within 12 months prior to the diagnosis of Hepatocellular Carcinoma for biomarkers and combinations derived from GALAD by using the relevant coefficients: GALAD=Gender+Age+AFP+AFP3%+AFP+DCP; GAA= Gender + Age + AFP; GAD=Gender + Age + DCP; GAAD=Gender + Age + AFP + DCP. **(A)** Receiver Operating Characteristics Curves. **(B)** Sensitivity for prediction of HCC within 1 – 36 months at 82% specificity.

Table 1.
Demographic information of patients by outcome status.

Variables	Cirrhosis Controls (n = 1123)	Hepatocellular Carcinoma (n=107)	Total (n=1230)
Sex (M:F) (%)	51.6:48.4	72.9:27.1	53.4:46.6
Median Age	60 (54:65)	62 (56:66)	60 (54:65)
Median BMI	30 (26.4:35.2)	32.1 (27.4:36.9)	30.2
Race: Ethnicity (%)			
Hispanic	9.2	10.3	9.3
Non-Hispanic Black	6.9	9.3	7.2
Non-Hispanic White	80.6	77.6	80.3
Other	3.2	2.8	3.3
Primary Etiology of Liver Disease (%)			
Hepatitis B	2.6	0	2.4
Hepatitis C	41.3	54.2	42.4
Alcohol	14.8	14	14.7
MASLD	26.6	22.4	26.3
Other	14.7	9.3	14.3
Median MELD Score	8.0 (7.0:11.0)	10 (8.0:12.0)	8.0 (7.0:11.0)
Child class (A:B:NA) (%)	75.1:24.8:0.1	61.7:37.4:0.9	73.9:25.9:0.2
Tumor Burden Stage (Very Early/Early/Late/)	NA	21/55/31/	NA

Interquartile range listed for continuous variables. BMI= body mass index. Tumor Burden Stage = Barcelona Clinic Liver Cancer staging simplified to use tumor number, size, and invasiveness but not performance status or Child class. NAFLD= non-alcoholic fatty liver disease.

Table 2 –
Performance characteristics rules derived from published cutoffs for AFP (t= 20 ng/mL) and GALAD (t= -1.36).

	AFP (t= 20 ng/mL)		GALAD (t= -1.36)		AFP (t= 6.45 ng/mL)*	
	Sensitivity (%)	Specificity (%)	Sensitivity (%)	Specificity (%)	Sensitivity (%)	Specificity (%)
All HCC						
3 months	24 (9,39)	97 (97,98)	62 (44,77)	82 (80,84)	41 (24,58)	82 (80,84)
6 months	19 (10,30)	97 (97,98)	65 (51,77)	82 (80,84)	45 (32,59)	82 (80,84)
12 months	12 (6,19)	97 (97,98)	62 (50,73)	82 (80,84)	42 (30,53)	82 (80,84)
Early HCC						
3 months	15 (4,31)	97 (97,98)	58 (36,76)	82 (80,84)	35 (16,52)	82 (80,84)
6 months	16 (7,28)	97 (97,98)	65 (50,79)	82 (80,84)	35 (16,52)	82 (80,84)
12 months	11 (5,19)	97 (97,98)	60 (46,72)	82 (80,84)	39 (26,53)	82 (80,84)
Viral (HCV)						
3 months	24 (6,47)	94 (93,96)	71 (47,88)	81 (77,84)	47 (24,71)	71 (67,75)
6 months	21 (9,38)	94 (93,96)	76 (58,91)	81 (77,84)	52 (34,71)	71 (67,75)
12 months	12 (5,21)	94 (93,96)	74 (59,87)	81 (77,84)	49 (33,66)	71 (67,75)
Non-Viral						
3 months	24 (6,44)	100 (100,100)	53 (25,76)	83 (80,86)	35 (12,59)	92 (90,94)
6 months	17 (6,33)	100 (100,100)	52 (32,70)	83 (80,86)	38 (20,57)	92 (90,94)
12 months	12 (2,22)	100 (100,100)	49 (30,66)	83 (80,86)	33 (18,49)	92 (90,94)

* AFP with cutoff = 6.25 ng/mL achieves the same specificity (82%) as GALAD with cutoff -1.36 which allows fair comparisons of sensitivity among all patients.

(95% CI) follows each performance metric.

Table 3

Area under the receiver operating curve (AUC) for sensitivities within 3,6, and 12 months of HCC diagnosis.

	AUC (95% CI)					
	AFP	DCP	GAA	GAD	GAAD	GALAD
All HCC						
3 months	0.69 (0.59,0.79)	0.78 (0.69,0.86)	0.73 (0.63,0.83)	0.79 (0.69,0.87)	0.8 (0.7,0.88)	0.80 (0.70,0.88)
6 months	0.71 (0.63,0.79)	0.74 (0.67,0.81)	0.74 (0.67,0.81)	0.74 (0.67,0.81)	0.8 (0.73,0.86)	0.81 (0.74,0.87)
12 months	0.66 (0.59,0.74)	0.71 (0.65,0.78)	0.71 (0.64,0.78)	0.74 (0.67,0.8)	0.76 (0.7,0.83)	0.78 (0.71,0.84)
Early HCC						
3 months	0.66 (0.54,0.78)	0.75 (0.64,0.84)	0.72 (0.6,0.82)	0.77 (0.65,0.87)	0.79 (0.67,0.88)	0.78 (0.67,0.88)
6 months	0.7 (0.61,0.78)	0.71 (0.63,0.78)	0.75 (0.68,0.82)	0.74 (0.66,0.82)	0.8 (0.73,0.87)	0.81 (0.74,0.87)
12 months	0.64 (0.55,0.72)	0.70 (0.62,0.78)	0.70 (0.62,0.77)	0.73 (0.64,0.8)	0.76 (0.67,0.83)	0.77 (0.68,0.84)
Viral						
3 months	0.67 (0.54,0.8)	0.81 (0.7,0.91)	0.79 (0.66,0.90)	0.85 (0.73,0.95)	0.83 (0.7,0.94)	0.83 (0.7,0.93)
6 months	0.70 (0.60,0.79)	0.81 (0.73,0.89)	0.74 (0.65,0.83)	0.78 (0.68,0.87)	0.83 (0.74,0.91)	0.84 (0.75,0.92)
12 months	0.65 (0.56,0.73)	0.78 (0.7,0.86)	0.71 (0.62,0.8)	0.78 (0.69,0.86)	0.8 (0.71,0.88)	0.81 (0.72,0.89)
Non-Viral						
3 months	0.71 (0.55,0.86)	0.76 (0.63,0.85)	0.70 (0.55,0.82)	0.72 (0.57,0.84)	0.77 (0.63,0.88)	0.78 (0.64,0.88)
6 months	0.71 (0.60,0.82)	0.69 (0.58,0.78)	0.73 (0.62,0.83)	0.71 (0.6,0.8)	0.77 (0.66,0.85)	0.78 (0.68,0.86)
12 months	0.66 (0.54,0.77)	0.66 (0.55,0.75)	0.70 (0.58,0.8)	0.69 (0.59,0.78)	0.73 (0.63,0.82)	0.74 (0.64,0.84)

GAA: Gender + Age + AFP; GAD: Gender + Age + DCP; GAAD: Gender + Age + AFP + DCP; GALAD: Gender+Age+AFPL3%+AFP+DCP; the coefficients from the full GALAD define the combinations of subsets of constituents (i.e. calculate the GALAD score with age and gender and use 0 for the constituents AFPL3%, DCP, or AFP not included in that particular combination.).

(95% CI) follows each performance metric

Table 4

Performance characteristics of tests using cutoffs that target high specificity or high sensitivity.

		Sensitivity (%) at 82% Specificity (95% CI)					
	n	AFP	DCP	GAA	GAD	GAAD	GALAD
All HCC							
3 months	34	41 (24,58)	66 (49,81)	47 (29,64)	66 (49,81)	62 (44,79)	62 (45,79)
6 months	57	45 (32,59)	52 (39,66)	45 (32,59)	54 (41,67)	58 (45,73)	65 (50,78)
12 months	69	42 (30,54)	50 (38,64)	44 (31,56)	52 (41,65)	57 (44,70)	62 (49,74)
Early HCC							
3 months	29	35 (16,52)	62 (40,80)	42 (24,64)	62 (40,80)	62 (40,80)	58 (36,77)
6 months	49	43 (29,58)	47 (31,63)	45 (29,59)	53 (38,69)	61 (46,77)	65 (49,81)
12 months	57	39 (26,54)	49 (34,64)	40 (26,54)	51 (37,66)	57 (43,72)	60 (43,74)
Viral							
3 months	18	41 (18,65)	67 (44,89)	59 (35,82)	78 (56,94)	71 (47,88)	71 (47,88)
6 months	30	39 (23,59)	68 (50,83)	48 (31,69)	65 (44,81)	67 (48,85)	73 (53,9)
12 months	36	35 (21,50)	64 (49,78)	49 (32,67)	64 (46,79)	67 (48,82)	68 (52,85)
Non-Viral							
3 months	16	47 (26,76)	59 (29,82)	41 (17,62)	47 (19,75)	59 (25,82)	59 (31,83)
6 months	27	45 (28,67)	52 (30,70)	48 (28,68)	41 (21,63)	52 (30,72)	59 (34,78)
12 months	33	39 (24,61)	41 (25,56)	45 (26,64)	39 (20,58)	49 (29,67)	53 (33,72)
		Specificity (%) at 90% Sensitivity (95% CI)					
	n	AFP	DCP	GAA	GAD	GAAD	GALAD
All HCC							
3 months	1123	24 (14,51)	35 (21,67)	14 (50,54)	40 (5,74)	35 (5,69)	40 (4,69)
6 months	1123	24 (16,42)	35 (17,54)	38 (13,56)	34 (19,56)	51 (18,67)	53 (18,67)
12 months	1123	18 (10,28)	25 (10,45)	18 (7,47)	24 (16,54)	35 (15,55)	35 (14,56)
Early HCC							
3 months	1123	21 (5,54)	25 (15,53)	14 (5,55)	18 (5,75)	35 (4,72)	40 (4,71)
6 months	1123	21 (14,42)	25 (7,52)	46 (6,58)	23 (6,58)	57 (6,69)	56 (5,71)
12 months	1123	16 (9,23)	25 (8,48)	18 (5,50)	22 (6,56)	30 (5,61)	34 (4,60)
Viral							
3 months	493	18 (10,51)	35 (10,51)	42 (1,71)	54 (3,84)	50 (1,75)	49 (1,73)
6 months	493	37 (12,54)	46 (12,54)	42 (22,63)	31 (18,62)	59 (35,75)	58 (41,82)
12 months	493	14 (7,46)	32 (7,46)	33 (1,50)	38 (4,61)	40 (1,67)	46 (1,68)
Non-Viral							
3 months	630	23 (7,47)	40 (9,79)	21 (8,59)	19 (6,73)	39 (7,7)	38 (7,69)
6 months	630	23 (8,46)	26 (4,46)	22 (9,62)	23 (7,56)	39 (8,69)	38 (8,69)
12 months	630	20 (5,32)	26 (4,42)	22 (9,54)	22 (8,56)	30 (8,61)	29 (8,60)

GAA: Gender + Age + AFP; GAD: Gender + Age + DCP; GAAD: Gender + Age + AFP + DCP; GALAD: Gender+Age+AFP+DCP

* The specificity varies with period prior to HCC diagnosis because the cutoffs to achieve 90% sensitivity for tests within different periods prior to HCC diagnosis differ. Cutoff selection for 82% specificity has uncertainty (Supplementary Table 9). (95% CI) follows each performance metric.

The 95% CI for sensitivity in this table reflect the uncertainty in both sensitivity and the cutoff to set the target specificity